1. **Insurance Risk Portfolio Analyzer**

Problem

An insurance company wants to analyze risk levels of customers.

You are given:

- n customers
- An array ages[]
- An array riskScores[] (0–100)

Rules:

- If age < 25 AND riskScore > 70 → "High Risk Youth"
- If age ≥ 60 → "Senior Risk"
- If riskScore ≥ 85 → "Very High Risk"
- Otherwise → "Normal Risk"

Tasks:

1. Classify each customer
2. Count number of customers in each category
3. Identify average risk score
4. Print index of highest risk customer

2. **Smart Loan Eligibility Checker**

Problem

A bank evaluates multiple applicants.

Inputs:

- creditScores[]
- monthlyIncomes[]
- existingLoans[] (number of loans)

Rules:

Reject if:

- creditScore < 600
- income < 25,000
- existingLoans ≥ 3

Otherwise:

- If creditScore ≥ 800 AND income > 1,00,000 → "Instant Approval"
- Else → "Standard Review"

Tasks:

- Print eligibility result for each applicant
- Count total approvals & rejections
- Find the applicant with best approval profile

3. **Electricity Usage Pattern Detector**

Problem

You are given daily electricity usage for 30 days:

int usage[]

Rules:

- Days with usage > 500 → "High Consumption"
- Days with usage < 100 → "Low Usage Alert"
- If 3 consecutive days > 500 → "Overload Warning"
- If average usage > 400 → "Heavy Month"

Tasks:

- Print classification for each day
- Detect if overload warning exists
- Calculate monthly average
- Count high consumption days

4. **Student Performance Heatmap**

Problem

You are given marks of students in 5 subjects:

int marks[n][5]

Rules:

- If any subject < 35 → "Fail"
- If average ≥ 85 → "Distinction"
- If average between 60–84 → "First Class"
- If average between 50–59 → "Second Class"

Tasks:

- Print result for each student
- Count number of distinctions
- Find subject with highest overall average

5. **Fraud Transaction Detector**

Problem

An e-commerce system tracks daily transactions:

double transactions[]

Rules:

- Transaction > 50,000 → Flag as suspicious
- If 2 suspicious transactions occur consecutively → "Potential Fraud"
- If average daily transaction > 40,000 → "High Value Account"

Tasks:

- Flag each suspicious transaction
- Detect fraud pattern
- Print suspicious transaction indices
- Calculate average transaction value